Handling Class Imbalance in Breast Ultrasound Image Classification

A Comparative Study on the BUSI Dataset using ResNet-34 and PyTorch

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Dataset:	BUSI [2]	**Platform:**	Kaggle (Python 3.12)

1. Introduction

Breast cancer is one of the most prevalent and life-threatening cancers worldwide, accounting for a significant fraction of cancer-related female mortality. Breast Ultrasound (BUS) imaging is a widely adopted, non-invasive diagnostic modality because of its low cost and absence of ionising radiation. However, automated multi-class classification of BUS images faces two key challenges: (i) subtle visual differences between benign and malignant lesions, and (ii) **class imbalance** — where training samples are unevenly distributed across classes.

In machine learning, class imbalance causes standard Cross-Entropy (CE) loss to bias the model toward the majority class. The resulting classifier may report high raw accuracy while completely failing to detect clinically critical minority classes such as *malignant* or *normal*. This is particularly dangerous in a medical context, where a missed malignancy can be fatal and incorrectly labelling a normal patient as diseased leads to unnecessary and costly intervention.

This report systematically evaluates **four mitigation strategies** on the BUSI dataset using a shared ResNet-34 backbone, with all training variables held fixed: (1) Baseline CE Loss, (2) Random Undersampling, (3) Oversampling via `WeightedRandomSampler`, and (4) Focal Loss combined with Data Augmentation. The study also provides a detailed justification for why SMOTE was deliberately excluded. Results show that Focal Loss + Augmentation yields the most balanced and clinically sound model.

2. Dataset Description and Class Imbalance

2.1 BUSI Dataset Overview

The **Breast Ultrasound Images (BUSI)** dataset [1] was collected at Badr University Hospital, Egypt, from 600 female patients aged 25–75. It contains grayscale ultrasound images organised into three clinical classes:

Table 1: Class Definitions in the BUSI Dataset

Class	Label	Clinical Description
Normal	0	No lesion present; healthy breast tissue with no detectable abnormality.
Benign	1	Non-cancerous lesion (e.g., cyst, fibroadenoma). Not immediately life-threatening.
Malignant	2	Cancerous lesion; highest clinical risk, requires urgent intervention.

Each image is paired with a segmentation mask file. Mask files (identified by `_mask` in their filename) were explicitly excluded during data loading. Class labels were derived from the parent directory name via dynamic `os.walk` traversal of the Kaggle input path.

2.2 Class Distribution and Imbalance

Table 2: Full Dataset Distribution

Class	Approx. Samples	Proportion	Ratio vs. Normal	Role
Normal	133	17.1%	1.0×	Minority
Benign	437	56.0%	3.3×	Dominant Majority
Malignant	210	26.9%	1.6×	Moderate Minority
Total	780	100%	—	—

Note on Imbalance: The *benign* class accounts for ∼56% of samples, creating a 3.3:1 imbalance over *normal*. A model trained with standard CE on this data will have gradients dominated by the benign class, producing high raw accuracy while dangerously under-detecting *normal* and *malignant* — both of which carry high clinical stakes.

2.3 Patient Holdout Split

A **stratified 70/15/15 split** was applied using `train_test_split` with `stratify=df['label']` and `random_state=42`, preserving class proportions across all partitions. The validation set is used exclusively for model selection (best checkpoint by minimum validation loss). The **test set** (117 images: 20 normal, 66 benign, 31 malignant) is held out and evaluated identically for all four experiments, ensuring a fair and reproducible comparison.

3. Methodology

3.1 Model Architecture and Shared Training Setup

Table 3: Shared Model and Training Hyperparameters (All Experiments)

Component	Configuration
Backbone	ResNet-34 [?], ImageNet pre-trained (all layers unfrozen)
Classification Head	`Dropout(p=0.5)` → `Linear(512, 3)`
Optimizer	Adam, learning rate $= 1 \times 10^{-4}$
Epochs / Batch Size	10 epochs, batch size 32
Checkpoint Strategy	Best validation loss saved via `deepcopy(model.state_dict())`
Input Resolution	224×224 pixels
Normalisation	ImageNet: μ=[0.485, 0.456, 0.406], σ=[0.229, 0.224, 0.225]

The custom `BUSIDataset` class (extending `torch.utils.data.Dataset`) handles image loading via PIL, RGB conversion, and configurable transform application. A reusable `train_experiment()` function manages the training loop and checkpoint logic; `evaluate_model()` generates classification reports on the held-out test set.

Data Augmentation (Experiment 4 training only): Additionally applies `RandomHorizontalFlip` `RandomRotation(15°)`, and `ColorJitter(brightness=0.1, contrast=0.1)` to expand effective training diversity and introduce realistic variation in image orientation and illumination.

3.2 Training and Validation Procedure

Each experiment initialises a extitfresh ResNet-34 model via `get_resnet_model()` so there is no weight sharing across experiments. The Adam optimizer is instantiated anew per experiment with `lr=1e-4`. During each of the 10 training epochs, the model switches between `train()` and `eval()` modes to correctly handle Dropout and Batch Normalisation. Validation loss is computed over the full validation set after every epoch; the lowest-loss checkpoint is saved with `deepcopy` and restored before test-set evaluation. This approach prevents the model from being evaluated at a coincidentally good or bad epoch and ensures reproducibility of reported metrics.

3.3 The Four Imbalance-Handling Strategies

Experiment 1 — Baseline (Standard Cross-Entropy): $\mathcal{L}_{CE} = -\frac{1}{N}\sum_i \log p(y_i \mid x_i)$. No imbalance correction; all classes contribute equally to the loss. Gradients are dominated by the most frequent class (*benign*). This is the lower-bound reference.

Experiment 2 — Random Undersampling: Training data is truncated so all classes have $n_{\min} = \min(n_c)$ samples, drawn without replacement using `groupby('label').sample(n=` `random_state=42)`. Standard CE loss applied on the resulting balanced but much smaller training set.

Experiment 3 — Oversampling (WeightedRandomSampler): Each sample receives weight $w_c = 1/n_c$ inversely proportional to class size. PyTorch's `WeightedRandomSampler` (`replacement=True`) produces balanced mini-batches while retaining all original training data. DataLoader `shuffle=False` since the sampler manages ordering. Standard

CE loss applied.

Experiment 4 — Focal Loss + Augmentation: Focal Loss [?] modifies CE by adding a modulating factor that suppresses well-classified (majority-class) examples and amplifies loss from misclassified (minority-class) examples:

$$\mathcal{L}_{FL} = -\frac{1}{N}\sum_{i} \alpha_{y_i}\,(1-p_i)^{\gamma}\,\log p_i, \quad \gamma = 2.0$$

Per-class weights: $\alpha_c = (1/n_c)\,/\,\sum_{c'}(1/n_{c'})$ (normalised inverse frequency). As $p_i \to 1$ (easy sample), $(1-p_i)^2 \to 0$, strongly suppressing its contribution. Combined with data augmentation for diversity and generalisation.

3.4 Why SMOTE Was Not Used

SMOTE [?] generates synthetic minority samples by linearly interpolating between real minority examples in raw pixel space. For 224×224 ultrasound images (150,528 dimensions per image), this produces blurry, physically implausible outputs that lack the speckle texture and lesion boundary morphology of genuine BUS scans. Breast ultrasound diagnostics rely heavily on these fine-grained characteristics; pixel-level SMOTE destroys them. `WeightedRandomSampler` achieves equivalent oversampling using only real images. Applying SMOTE meaningfully to image data requires encoding images into a compact latent space before interpolation — a complex generative pipeline well beyond the scope of this assignment and unnecessary given the superior alternatives employed here.

4. Results on All Four Experiments

All models are evaluated on the same 117-image test set. Metric values are taken strictly from the classification report output screenshots.

Table 4: Experiment 1 — Baseline (Standard CE) **Overall Accuracy: 0.79**

Class	Precision	Recall	F1-Score	Support
Normal	0.83	0.50	0.62	20
Benign	0.78	0.94	0.86	66
Malignant	0.81	0.68	0.74	31
Macro Avg	0.81	0.71	0.74	117
Weighted Avg	0.80	0.79	0.78	117

Table 5: Experiment 2 — Undersampling **Overall Accuracy: 0.79**

Class	Precision	Recall	F1-Score	Support
Normal	0.65	0.85	0.74	20
Benign	**0.96**	0.74	0.84	66
Malignant	0.68	**0.87**	0.76	31
Macro Avg	0.76	0.82	0.78	117
Weighted Avg	0.83	0.79	0.80	117

Table 6: Experiment 3 — Oversampling (WeightedRandomSampler) **Overall Accuracy: 0.85**

Class	Precision	Recall	F1-Score	Support
Normal	**1.00**	0.55	0.71	20
Benign	0.83	**0.97**	**0.90**	66
Malignant	0.86	0.81	0.83	31
Macro Avg	**0.90**	0.78	0.81	117
Weighted Avg	0.87	0.85	0.85	117

Table 7: Experiment 4 — Focal Loss + Data Augmentation **Overall Accuracy: 0.85**

Class	Precision	Recall	F1-Score	Support
Normal	0.82	**0.90**	**0.86**	20
Benign	**0.89**	0.86	0.88	66
Malignant	0.81	0.81	**0.81**	31
Macro Avg	0.84	**0.86**	**0.85**	117
Weighted Avg	0.86	0.85	**0.85**	117

5. Comparative Analysis and Discussion

5.1 Head-to-Head Summary Tables

Table 8: F1-Score and Accuracy Comparison (best green, worst red)

Method	Nor. F1	Ben. F1	Mal. F1	Macro F1	Wtd. F1	Acc.
Baseline (CE)	0.62	0.86	0.74	0.74	0.78	0.79
Undersampling	0.74	0.84	0.76	0.78	0.80	0.79
Oversampling (WRS)	0.71	**0.90**	0.83	0.81	0.85	**0.85**
Focal + Aug	**0.86**	0.88	**0.81**	**0.85**	**0.85**	**0.85**

Table 9: Per-Class Recall and Precision Comparison

Method	Nor. Rec.	Ben. Rec.	Mal. Rec.	Nor. Prec.	Ben. Prec.	Mal. Prec.
Baseline	0.50	0.94	0.68	0.83	0.78	0.81
Undersampling	0.85	0.74	**0.87**	0.65	**0.96**	0.68
Oversampling	0.55	**0.97**	0.81	**1.00**	0.83	0.86
Focal + Aug	**0.90**	0.86	0.81	0.82	**0.89**	0.81

5.2 Key Observations and Reasoning

1. **Baseline confirms the majority-class bias.** Despite 79% accuracy, the model misses one in every two healthy (normal) patients (recall = 0.50). It heavily favours *benign* (recall 0.94), confirming that unweighted CE on imbalanced data produces a clinically unsafe classifier with a macro F1 of only 0.74.
2. **Undersampling: balanced recall, lower precision, no accuracy gain.** Equalising class sizes forces the model to attend to all classes equally: normal recall rises to 0.85 and malignant to 0.87. However, discarding ∼75% of benign training images causes information loss, dropping precision to 0.65/0.68 for minority classes, generating more false positives. Accuracy stays at 79%, showing the clear cost of data wastage.
3. **Oversampling: accuracy gain but precision-recall trade-off.** Maintaining the full dataset while rebalancing sampling distribution pushes accuracy to 85%. Normal class precision is a perfect **1.00** — every normal prediction is correct — but recall is only 0.55, meaning 45% of actual normal cases are missed. This is a characteristic oversampling effect: repeated exposure to minority images causes the model to become highly selective rather than sensitive.
4. **Focal Loss + Augmentation: best all-round performance.** Achieves the highest macro F1 (**0.85**) with the most balanced per-class profile — no recall falls below 0.81. The $(1 - p_t)^2$ modulation forces larger gradients from hard, misclassified (minority-class) examples, while α_c weighting further penalises minority-class errors. Augmentation prevents overfitting to the repeated minority patterns that Focal Loss highlights. This is the only strategy achieving high accuracy *and* balanced recall simultaneously.
5. **Accuracy is an insufficient metric for imbalanced data.** Baseline and Undersampling both report 79% accuracy despite drastically different clinical utility.

Oversampling and Focal+Aug both reach 85%, yet differ by 4 macro F1 points (0.81 vs. 0.85). **Macro F1 must be the primary evaluation criterion** for imbalanced medical classification — it weights all classes equally and is insensitive to majority-class dominance.

6. Conclusion

This study conducted a controlled four-way comparison of class imbalance mitigation strategies on the BUSI breast ultrasound dataset using ResNet-34. All hyperparameters were held constant so that performance differences are attributable solely to each strategy.

Ranked performance (Macro F1):

$$\underbrace{\text{Focal Loss + Aug}}_{0.85} > \underbrace{\text{Oversampling (WRS)}}_{0.81} > \underbrace{\text{Undersampling}}_{0.78} > \underbrace{\text{Baseline (CE)}}_{0.74}$$

Recommendation: Focal Loss + Augmentation achieves **85% accuracy** and the highest **macro F1 = 0.85**, with the most balanced recall: Normal 0.90, Benign 0.86, Malignant 0.81. It addresses imbalance at the loss level without discarding data or generating synthetic images, and is the most principled, clinically sound approach for this task.

Each strategy exposes a distinct trade-off: undersampling sacrifices data and reduces precision; oversampling improves accuracy but leaves minority recall limited; only Focal Loss with augmentation achieves both clinical safety and high overall accuracy. Future directions include combining Focal Loss with `WeightedRandomSampler`, extended training with cosine annealing, EfficientNet/ViT backbones, and incorporating segmentation masks as spatial attention guidance.

References

[1] W. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, "Dataset of breast ultrasound images," *Data in Brief*, vol. 28, p. 104863, 2020. https://doi.org/10.1016/j.dib.2019.104863

[2] A. Shah, "Breast Ultrasound Images Dataset," Kaggle, 2021. https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset